DATA STRUCTURE

LAB PROGRAMS

20 . Merge Sort

```
#include <stdio.h>
void merge(int arr[], int l, int m, int r) {
   int i=l,j=m+1,k=0,temp[100];
   while(i<=m && j<=r) {
      if(arr[i] < arr[j])
         temp[k++] = arr[i++];
      else
         temp[k++] = arr[j++];
   }
   while(i<=m)
      temp[k++] = arr[i++];
   while(j<=r)
      temp[k++] = arr[j++];
   for(i=l,k=0;i<=r;i++,k++)
      arr[i] = temp[k];
}
void mergeSort(int arr[], int l, int r) {
   if(l<r) {
      int m=(l+r)/2;
      mergeSort(arr,l,m);
      mergeSort(arr,m+1,r);
      merge(arr,l,m,r);
   }
}
```

```c
int main() {
    int arr[] = {38,27,43,3,9,82,10};
    int n = 7;
    mergeSort(arr,0,n-1);
    printf("Sorted Array:\n");
    for(int i=0;i<n;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Output

```
Sorted Array:
3 9 10 27 38 43 82

=== Code Execution Successful ===
```